Ivan Valbusa

The fontsize package

v0.8.5 – 2021/08/04

The serif typeface is Cochineal, created by Michael Sharpe on the basis of Sebastian Kosch's Crimson. The `typewriter` font is Inconsolata, designed by Raph Levien and the sans serif text is typeset in Linux Biolinum, created by Philipp H. Poll.

Legalese

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Feedback

If you have any questions, feedback or requests please email me at `ivan dot valbusa at gmail dot com`. If you need specific features not already implemented, remember to attach the example files.

User's guide to fontsize

A small package to change the font size

v0.8.5 – 2021/08/04

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Abstract

The fontsize package allows you to set arbitrary sizes for the main font of the document and to adjust the line spacing accordingly. It also provides new font sizing commands and lets you generate your own .clo file for better tuning.

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Credits

The code is an excerpt of the scrextend and scrfontsizes packages (Neukam et al. 2020), with few minor changes to make it completely independent of the KOMA-script bundle and with a couple of new features. The package is updated to version 3390 2019-11-18 17:09:35Z of the scrkernel-fonts.dtx file.

Caveat

Although effective with any value of the font size (as long as the font is scalable), the package is meant for small font size changes. For truly optimal results you should (re)define your own .clo file. In fact note that the file generated with \generateclofile, as well as the macros redefined via \changefontsize, are a subset of the standard .clo files.

1 Usage

The package can be loaded, as usual, with:

```
\usepackage[⟨options⟩]{fontsize}
```

1.1 Options

`fontsize` = ⟨*fontsize*⟩ default: `10pt`

Sets the normal size to ⟨*fontsize*⟩ and the normal line spacing to 1.2⟨*fontsize*⟩. The other sizes (`\tiny`, `\scriptsize`, ecc.) and line spacings are set proportionally. You can use any unit recognized by TEX and you can omit the unit as well (in which case it defaults to `pt`).

Note that `fontsize` does not redefine some commands of the standard *.clo* files such as those related to the page layout and few others. For example the `12pt` class option redefine `\Huge` (via `\let\Huge=\huge`) which is not redefined by `fontsize=12pt`. For these reasons, if possible, you should always set a font size class option that comes close to the one loaded by the package. For example with `fontsize=12.2pt` you should load the `12pt` class option.

`parindent` = ⟨*dimen*⟩ default: `1.5`⟨*fontsize*⟩

Sets the value of `\parindent`. You can use any unit recognized by TEX but for consistency it is preferable to use a font-dependent unit such as em.

Note that the "em" is simply an arbitrary unit that comes with a particular font. The value of `\parindent` given in ems depends on both the font used and on how or where it is set: whether before or after `\begin{document}`. For this reason, don't expect 1 em to always be equivalent to the currently specified ⟨*fontsize*⟩. If you don't like this behavior, use absolute units such as pt, cm or mm.

`cloname` = ⟨*name*⟩ default: `fontsize`

If declared *before* `fontsize`, it loads the file named ⟨*name*⟩⟨*fontsize*⟩`.clo` to setup the font sizes. This file can be generated through the `\generateclofile` command (see below).

1.2 Commands

`\changefontsize[`⟨*baselineskip*⟩`]{`⟨*fontsize*⟩`}` iff ⟨*baselineskip*⟩ ≥⟨*fontsize*⟩

Sets the normal size to ⟨*fontsize*⟩ and the normal line spacing to ⟨*baselineskip*⟩. The other sizes (`\tiny`, `\scriptsize`, ecc.) and line spacings are set proportionally. If the first optional argument is omitted, the command has the same effect of the `fontsize` option. You can use any unit recognized by TEX and you can omit the unit as well (in which case it defaults to `pt`).

```
\changefontsize[13.5pt]{11.5pt}
```

`\generateclofile{`⟨*name*⟩`}[`⟨*baselineskip*⟩`]{`⟨*fontsize*⟩`}` iff ⟨*baselineskip*⟩ ≥⟨*fontsize*⟩

Generates a file named ⟨*name*⟩⟨*fontsize*⟩`.clo` in which the normal size is ⟨*fontsize*⟩ and the normal line spacing is ⟨*baselineskip*⟩. The other sizes (`\tiny`, `\scriptsize`, `\small`, ecc.) and line spacings are set proportionally. You can use any unit recognized by TEX and you can omit the unit as well (in which case it defaults to `pt`). If `cloname=`⟨*name*⟩ *and* `fontsize=`⟨*fontsize*⟩, the generated file will be automatically used to setup the main sizes.

In fact the `.clo` *file is only needed if you intend to modify it. In this case remember to comment the command, otherwise the edited file will be overwritten.*

```
\usepackage[cloname=fs,fontsize=11.5]{fontsize}    % loads fs11.5.clo
 \generateclofile{fs}[13.5]{11.5}                  % creates fs11.5.clo
```

`\printsamples{`⟨*baselineskip*⟩`}{`⟨*fontsize*⟩`}[`⟨*sizes*⟩`]` iff ⟨*baselineskip*⟩ ≥ ⟨*fontsize*⟩

A command thought to test the `fontsize` settings. It prints a text sample for every size given in the third optional argument. If this argument is omitted, it prints a sample for each available size. See the examples on page 8.

```
\printsamples{13pt}{11pt}[normalsizer,normalsizerr,normalsizerrr]
\printsamples{13.5pt}{12pt}
```

`\sampletext{`⟨*text*⟩`}`

Defines the ⟨*text*⟩ to be used by `\printsamples`.

1.3 New font sizing commands

The package provides three additional font sizing commands for each standard size (plus the new `\HUGE` command). See table 1 for details.

\tiny	Lipsum
\tinyr	Lipsum
\tinyrr	Lipsum
\tinyrrr	Lipsum
\scriptsize	Lipsum
\scriptsizer	Lipsum
\scriptsizerr	Lipsum
\scriptsizerrr	Lipsum
\footnotesize	Lipsum
\footnotesizer	Lipsum
\footnotesizerr	Lipsum
\footnotesizerrr	Lipsum
\small	Lipsum
\smallr	Lipsum
\smallrr	Lipsum
\smallrrr	Lipsum
\normalsize	Lipsum
\normalsizer	Lipsum
\normalsizerr	Lipsum
\normalsizerrr	Lipsum
\large	Lipsum
\larger	Lipsum
\largerr	Lipsum
\largerrr	Lipsum
\Large	Lipsum
\Larger	Lipsum
\Largerr	Lipsum
\Largerrr	Lipsum
\LARGE	Lipsum
\LARGEr	Lipsum
\LARGErr	Lipsum
\LARGErrr	Lipsum
\huge	Lipsum
\huger	Lipsum
\hugerr	Lipsum
\hugerrr	Lipsum
\Huge	Lipsum
\Huger	Lipsum
\Hugerr	Lipsum
\Hugerrr	Lipsum
\HUGE	Lipsum
\HUGEr	Lipsum
\HUGErr	Lipsum
\HUGErrr	Lipsum

If you want these commands to work as expected, the font must be smoothly scalable. This is the case of many Type 1 and Open Type fonts such as Cochineal used in this document (Sharpe 2020), but if the font has optical sizes, unusual things can happen: some larger sizes may appear smaller. For example, with the Latin Modern typeface (Jackowski and Nowacki 2009) a similar situation can be encountered when normal size is set to 12 pt:

`\largerr`	15.84 pt	Hello world
`\larger`	15.12 pt	Hello world
`\large`	14.40 pt	Hello world

In fact the font used for the `\larger` 'Hello world' is bigger than that used for the `\large` one. This optical effect is due to the minor thickness of the stems, which makes the words appear shorter and clearer, but not less high (as you can see in the figure on the right which shows the letter H appropriately scaled to see the differences better).

Table 1: The value of all available sizes (rounded to the second decimal place) relative to `\normalsize` ranging from 8 pt to 12 pt with increments of 0.5 pt. The first row of each size shows the font size; the second row (colored in gray) shows the default line spacing

`\tiny`	4.00	4.25	4.50	4.75	5.00	5.25	5.50	5.75	6.00
	4.80	5.06	5.40	5.65	6.00	6.30	6.55	6.90	7.14
`\tinyr`	4.33	4.60	4.88	5.15	5.42	5.69	5.96	6.23	6.50
	5.20	5.48	5.85	6.12	6.50	6.83	7.09	7.48	7.74
`\tinyrr`	4.67	4.96	5.25	5.54	5.83	6.12	6.42	6.71	7.00
	5.60	5.90	6.30	6.59	7.00	7.35	7.64	8.05	8.33
`\tinyrrr`	5.00	5.31	5.62	5.94	6.25	6.56	6.88	7.19	7.50
	6.00	6.32	6.75	7.07	7.50	7.87	8.18	8.62	8.93
`\scriptsize`	5.33	5.67	6.00	6.33	6.67	7.00	7.33	7.67	8.00
	6.40	6.74	7.20	7.54	8.00	8.40	8.73	9.20	9.52
`\scriptsizer`	5.67	6.02	6.38	6.73	7.08	7.44	7.79	8.15	8.50
	6.80	7.16	7.65	8.01	8.50	8.93	9.27	9.78	10.12
`\scriptsizerr`	6.00	6.38	6.75	7.12	7.50	7.88	8.25	8.62	9.00
	7.20	7.59	8.10	8.48	9.00	9.45	9.82	10.35	10.71
`\scriptsizerrr`	6.33	6.73	7.13	7.52	7.92	8.31	8.71	9.10	9.50
	7.60	8.01	8.55	8.95	9.50	9.98	10.36	10.93	11.31
`\footnotesize`	6.67	7.08	7.50	7.92	8.33	8.75	9.17	9.58	10.00
	8.00	8.43	9.00	9.42	10.00	10.50	10.91	11.50	11.90
`\footnotesizer`	6.83	7.25	7.68	8.10	8.53	8.96	9.38	9.81	10.24
	8.19	8.63	9.21	9.64	10.24	10.75	11.17	11.77	12.18
`\footnotesizerr`	6.98	7.42	7.86	8.29	8.73	9.17	9.60	10.04	10.48
	8.38	8.83	9.43	9.87	10.48	11.00	11.43	12.05	12.47
`\footnotesizerrr`	7.14	7.59	8.03	8.48	8.93	9.37	9.82	10.27	10.71
	8.57	9.03	9.64	10.09	10.71	11.25	11.69	12.32	12.75
`\small`	7.30	7.76	8.21	8.67	9.13	9.58	10.04	10.49	10.95
	8.76	9.23	9.86	10.32	10.95	11.50	11.94	12.59	13.03

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\smallr	7.47	7.94	8.41	8.88	9.34	9.81	10.28	10.75	11.21
	8.97	9.45	10.09	10.56	11.21	11.77	12.23	12.89	13.34
\smallrr	7.65	8.13	8.61	9.08	9.56	10.04	10.52	11.00	11.48
	9.18	9.67	10.33	10.81	11.48	12.05	12.52	13.20	13.66
\smallrrr	7.82	8.31	8.80	9.29	9.78	10.27	10.76	11.25	11.74
	9.39	9.89	10.56	11.06	11.74	12.32	12.80	13.50	13.97
\normalsize	8.00	8.50	9.00	9.50	10.00	10.50	11.00	11.50	12.00
	9.60	10.20	10.80	11.40	12.00	12.60	13.20	13.80	14.40
\normalsizer	8.40	8.93	9.45	9.98	10.50	11.03	11.55	12.08	12.60
	10.08	10.62	11.34	11.87	12.60	13.23	13.74	14.49	14.99
\normalsizerr	8.80	9.35	9.90	10.45	11.00	11.55	12.10	12.65	13.20
	10.56	11.13	11.88	12.44	13.20	13.86	14.40	15.18	15.71
\normalsizerrr	9.20	9.77	10.35	10.92	11.50	12.07	12.65	13.22	13.80
	11.04	11.63	12.42	13.00	13.80	14.49	15.05	15.87	16.42
\large	9.60	10.20	10.80	11.40	12.00	12.60	13.20	13.80	14.40
	11.52	12.14	12.96	13.57	14.40	15.12	15.71	16.56	17.14
\larger	10.08	10.71	11.34	11.97	12.60	13.23	13.86	14.49	15.12
	12.10	12.74	13.61	14.24	15.12	15.88	16.49	17.39	17.99
\largerr	10.56	11.22	11.88	12.54	13.20	13.86	14.52	15.18	15.84
	12.67	13.35	14.26	14.92	15.84	16.63	17.28	18.22	18.85
\largerrr	11.04	11.73	12.42	13.11	13.80	14.49	15.18	15.87	16.56
	13.25	13.96	14.90	15.60	16.56	17.39	18.06	19.04	19.71
\Large	11.52	12.24	12.96	13.68	14.40	15.12	15.84	16.56	17.28
	13.82	14.57	15.55	16.28	17.28	18.14	18.85	19.87	20.56
\Larger	12.10	12.85	13.61	14.36	15.12	15.88	16.63	17.39	18.14
	14.52	15.29	16.33	17.09	18.14	19.05	19.79	20.87	21.59
\Largerr	12.67	13.46	14.26	15.05	15.84	16.63	17.42	18.22	19.01
	15.21	16.02	17.11	17.91	19.01	19.96	20.73	21.86	22.62
\Largerrr	13.25	14.08	14.90	15.73	16.56	17.39	18.22	19.04	19.87
	15.90	16.75	17.88	18.72	19.87	20.87	21.68	22.85	23.65
\LARGE	13.82	14.69	15.55	16.42	17.28	18.14	19.01	19.87	20.74
	16.59	17.48	18.66	19.54	20.74	21.77	22.62	23.85	24.68
\LARGEr	14.52	15.42	16.33	17.24	18.14	19.05	19.96	20.87	21.77
	17.42	18.35	19.60	20.51	21.77	22.86	23.75	25.04	25.91
\LARGErr	15.21	16.16	17.11	18.06	19.01	19.96	20.91	21.86	22.81
	18.25	19.23	20.53	21.49	22.81	23.95	24.88	26.23	27.15
\LARGErrr	15.90	16.89	17.89	18.88	19.88	20.87	21.86	22.86	23.85
	19.08	20.10	21.46	22.47	23.85	25.04	26.02	27.43	28.38
\huge	16.59	17.63	18.67	19.70	20.74	21.78	22.81	23.85	24.89
	19.91	20.98	22.40	23.45	24.89	26.13	27.15	28.62	29.62
\huger	17.42	18.51	19.60	20.69	21.78	22.86	23.95	25.04	26.13
	20.90	22.03	23.52	24.62	26.13	27.44	28.50	30.05	31.09

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\hugerr	18.25	19.39	20.53	21.67	22.81	23.95	25.09	26.23	27.37
	21.90	23.07	24.63	25.79	27.37	28.74	29.86	31.48	32.57
\hugerrr	19.08	20.27	21.46	22.65	23.85	25.04	26.23	27.42	28.61
	22.89	24.12	25.75	26.96	28.61	30.04	31.21	32.91	34.05
\Huge	19.90	21.15	22.39	23.64	24.88	26.12	27.37	28.61	29.86
	23.88	25.17	26.87	28.13	29.86	31.35	32.57	34.33	35.53
\Huger	20.89	22.19	23.50	24.80	26.11	27.42	28.72	30.03	31.33
	25.07	26.41	28.20	29.52	31.33	32.90	34.18	36.03	37.29
\Hugerr	21.87	23.24	24.61	25.97	27.34	28.71	30.07	31.44	32.81
	26.25	27.65	29.53	30.91	32.81	34.45	35.79	37.73	39.04
\Hugerrr	22.86	24.28	25.71	27.14	28.57	30.00	31.43	32.86	34.28
	27.43	28.90	30.86	32.30	34.28	36.00	37.40	39.43	40.80
\HUGE	24.05	25.55	27.05	28.56	30.06	31.56	33.07	34.57	36.07
	28.86	30.41	32.46	33.98	36.07	37.88	39.35	41.48	42.93
\HUGEr	25.24	26.82	28.39	29.97	31.55	33.13	34.70	36.28	37.86
	30.29	31.91	34.07	35.67	37.86	39.75	41.30	43.54	45.05
\HUGErr	26.43	28.08	29.74	31.39	33.04	34.69	36.34	38.00	39.65
	31.72	33.42	35.68	37.35	39.65	41.63	43.25	45.60	47.18
\HUGErrr	27.62	29.35	31.08	32.80	34.53	36.26	37.98	39.71	41.44
	33.15	34.93	37.29	39.04	41.44	43.51	45.20	47.65	49.31

1.4 Some text samples

The following samples are obtained with the `\printsamples` command (see section 1.2):

```
\printsamples{12.6pt}{10.6pt}[normalsize,normalsizer,normalsizerr,
  normalsizerrr,large,larger,largerr,largerrr]
```

Text samples for \normalsize 10.6pt on 12.6pt

`\normalsize`: 10.6pt on 12.6pt. As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena.

`\normalsizer`: 11.1pt on 13.1pt. As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena.

`\normalsizerr`: 11.7pt on 13.8pt. As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of

practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena.

`\normalsizerrr`: 12.2pt on 14.4pt. As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena.

`\large`: 12.7pt on 15pt. As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena.

`\larger`: 13.4pt on 15.8pt. As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena.

`\largerr`: 14pt on 16.5pt. As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena.

`\largerrr`: 14.6pt on 17.3pt. As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena.

2 Known issues

The `parindent` option does not work with classes and packages which sets `\parindent` inside the `\AtBeginDocument` hook (and other hooks). Among these is the `tufte` class (Kleb et al. 2015). In such cases you can set the default indentation, for example, with `\setlength{parindent}{`⟨*dimen*⟩`}` declared *after* `\begin{document}`.

3 The Code

The parts added to the basic code of `scrextend` and `scrfontsizes` are highlighted in blue.

All the options of the package are defined via `xkeyval` (Adriaens 2014), while the `xfp` package (The LaTeX Project 2021) provides the `\fpeval` command to use instead of `\dimexpr` for some size calculations. Only two new package options, called `cloname` and `parindent` are defined.

```
\@ifpackageloaded{xkeyval}{}{\RequirePackage{xkeyval}}
\@ifpackageloaded{xfp}{}{\RequirePackage{xfp}}
\DeclareOptionX{cloname}[fontsize]{\def\@fontsizefilebase{#1}}
\DeclareOptionX{parindent}[1.5\@tempdimb]{\def\@fontsizeparindent{#1}}
```

Excerpt from `scrextend.sty` Here the `scrextend` code begins. All the KOMA related codes have been removed and only a few names have been changed. The value of `\@fontsizefilebase` is set to `fontsize` instead of `scrsize`.

```
\DeclareOptionX{fontsize}[10]{%
  \expandafter\@defaultunits\expandafter\@tempdima#1 pt\relax\@nnil
  \edef\@tempa{#1}%
  \setlength{\@tempdimb}{\@tempdima}%
  \edef\@tempb{\strip@pt\@tempdimb}%
  \addtolength{\@tempdimb}{-10\p@}%
  \edef\@ptsize{\strip@pt\@tempdimb}%
  \edef\@tempa{%
    \noexpand\makeatletter
      \noexpand\InputIfFileExists{\@fontsizefilebase\@tempa.clo}{%
   \noexpand\PackageInfo{fontsize}{%
     File `\@fontsizefilebase\@tempa.clo' used to setup font sizes}%
   }{%
   \noexpand\InputIfFileExists{\@fontsizefilebase\@tempb pt.clo}{%
    \noexpand\PackageInfo{fontsize}{%
     File `\@fontsizefilebase\@tempb pt.clo' used instead of%
   \noexpand\MessageBreak
     file `\@fontsizefilebase\@tempa.clo' to setup font sizes}%
   }{%
    \noexpand\changefontsize{#1}%
   }%
  }%
 \noexpand\catcode`\noexpand\@=\the\catcode`\@
 }%
  \expandafter\AtEndOfPackage\expandafter{\@tempa}%
```

```
}
```

Now we can execute and process the new options defined above. As `cloname` is responsible for defining `\@fontsizefilebase` it must precede `fontsize` which uses this command. The `\ctm@setlength` and `\changefontsize` commands are simply new names, respectively, for `\scr@setlength` and `\changefontsizes` used in `scrextend.sty`.

```
\ExecuteOptionsX{cloname,fontsize,parindent}
\ProcessOptionsX*\relax

\newcommand*\ctm@setlength[4]{%
  \ifdim #2<\ifdim #2=\z@ \z@ \else \p@\fi
  \ifdim #3<\ifdim #3=\z@ \z@ \else \p@\fi
  \ifdim #4<\ifdim #4=\z@ \z@ \else \p@\fi
  \setlength{#1}{\p@ \@plus\p@ \@minus\p@}%
  \else
  \setlength{#1}{\p@ \@plus\p@ \@minus#4}%
  \fi
  \else
  \ifdim #4<\ifdim #4=\z@ \z@ \else \p@\fi
  \setlength{#1}{\p@ \@plus#3 \@minus\p@}%
  \else
  \setlength{#1}{\p@ \@plus#3 \@minus#4}%
  \fi
  \fi
  \else
  \ifdim #3<\ifdim #3=\z@ \z@ \else \p@\fi
  \ifdim #4<\ifdim #4=\z@ \z@ \else \p@\fi
  \setlength{#1}{#2 \@plus\p@ \@minus\p@}%
  \else
  \setlength{#1}{#2 \@plus\p@ \@minus#4}%
  \fi
  \else
  \ifdim #4<\ifdim #4=\z@ \z@ \else \p@\fi
  \setlength{#1}{#2 \@plus#3 \@minus\p@}%
  \else
  \setlength{#1}{#2 \@plus#3 \@minus#4}%
  \fi
  \fi
  \fi
}
\providecommand*{\changefontsize}[2][1.2\@tempdima]{%
  \@defaultunits\@tempdima#2pt\relax\@nnil
  \setlength{\@tempdimc}{\@tempdima}%
  \addtolength{\@tempdimc}{-10\p@}%
  \edef\@ptsize{\strip@pt\@tempdimc}%
  \@defaultunits\@tempdimb#1pt\relax\@nnil
```

The value of `\@tempdimc` set in `scrextend.sty`

```
\setlength{\@tempdimc}{\dimexpr (100\@tempdimb / \@tempdima * \p@)}
```

returns an error when the the line spacing is greater than 163.84 pt or the value passed to the `fontsize` option is grater than 136.53 pt. This is due to the intermediate calculation: 100\@tempdimb = 16 384 pt, which is the maximum dimension that may be stored in plain TEX and LATEX. The problem, which is probably more theoretical than practical, can be solved simply by replacing the above code with this (thanks to Claudio Beccari):

```
\setlength{\@tempdimc}{\fpeval{100*\@tempdimb/\@tempdima}pt}%

\edef\@tempb{\the\@tempdimc}%
\setlength{\@tempdimc}{\@tempdima}%
\def\@tempa##1##2##3##4##5\@nnil{\def##1{##2.##3##4}}%
\expandafter\@tempa\expandafter\@tempb\@tempb\@nnil%
\ctm@setlength{\abovedisplayskip}%
{.8333\@tempdimb}{.1667\@tempdimb}{.5\@tempdimb}%
\ctm@setlength{\abovedisplayshortskip}{\z@}{.25\@tempdimb}{\z@}%
\ctm@setlength{\belowdisplayshortskip}%
{.5\@tempdimb}{.25\@tempdimb}{.25\@tempdimb}%
\setlength{\belowdisplayskip}{\abovedisplayskip}%
\edef\normalsize{%
  \noexpand\@setfontsize\noexpand\normalsize
  {\the\@tempdima}{\the\@tempdimb}%
  \abovedisplayskip \the\abovedisplayskip
  \abovedisplayshortskip \the\abovedisplayshortskip
  \belowdisplayskip \the\belowdisplayskip
  \belowdisplayshortskip \the\belowdisplayshortskip
  \let\noexpand\@listi\noexpand\@listI
}%
\setlength{\@tempdima}{0.9125\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\ctm@setlength{\abovedisplayskip}%
{.8333\@tempdimb}{.1667\@tempdimb}{.5\@tempdimb}%
\ctm@setlength{\abovedisplayshortskip}{\z@}{.25\@tempdimb}{\z@}%
\ctm@setlength{\belowdisplayshortskip}%
{.5\@tempdimb}{.25\@tempdimb}{.25\@tempdimb}%
\setlength{\belowdisplayskip}{\abovedisplayskip}%
\edef\small{%
  \noexpand\@setfontsize\noexpand\small
  {\the\@tempdima}{\the\@tempdimb}%
  \abovedisplayskip \the\abovedisplayskip
  \abovedisplayshortskip \the\abovedisplayshortskip
  \belowdisplayskip \the\belowdisplayskip
  \belowdisplayshortskip \the\belowdisplayshortskip
  \let\noexpand\@listi\noexpand\@listi@small
}%
\setlength{\@tempdima}{.83334\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\ctm@setlength{\abovedisplayskip}%
{.8333\@tempdimb}{.1667\@tempdimb}{.5\@tempdimb}%
\ctm@setlength{\abovedisplayshortskip}{\z@}{.25\@tempdimb}{\z@}%
\ctm@setlength{\belowdisplayshortskip}%
{.5\@tempdimb}{.25\@tempdimb}{.25\@tempdimb}%
```

```
  \setlength{\belowdisplayskip}{\abovedisplayskip}%
\edef\footnotesize{%
    \noexpand\@setfontsize\noexpand\footnotesize
    {\the\@tempdima}{\the\@tempdimb}%
    \abovedisplayskip \the\abovedisplayskip
    \abovedisplayshortskip \the\abovedisplayshortskip
    \belowdisplayskip \the\belowdisplayskip
    \belowdisplayshortskip \the\belowdisplayshortskip
    \let\noexpand\@listi\noexpand\@listi@footnotesize
  }%
  \setlength{\@tempdima}{.66667\@tempdimc}%
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\scriptsize{%
    \noexpand\@setfontsize\noexpand\scriptsize
    {\the\@tempdima}{\the\@tempdimb}%
  }%
  \setlength{\@tempdima}{.5\@tempdimc}%
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\tiny{%
    \noexpand\@setfontsize\noexpand\tiny
    {\the\@tempdima}{\the\@tempdimb}%
  }%
  \setlength{\@tempdima}{1.2\@tempdimc}%
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\large{%
    \noexpand\@setfontsize\noexpand\large
    {\the\@tempdima}{\the\@tempdimb}%
  }%
  \setlength{\@tempdima}{1.44\@tempdimc}%
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\Large{%
    \noexpand\@setfontsize\noexpand\Large
    {\the\@tempdima}{\the\@tempdimb}%
  }%
  \setlength{\@tempdima}{1.728\@tempdimc}%
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\LARGE{%
    \noexpand\@setfontsize\noexpand\LARGE
    {\the\@tempdima}{\the\@tempdimb}%
  }%
  \setlength{\@tempdima}{2.074\@tempdimc}%
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\huge{%
    \noexpand\@setfontsize\noexpand\huge
    {\the\@tempdima}{\the\@tempdimb}%
  }%
  \setlength{\@tempdima}{2.488\@tempdimc}%
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\Huge{%
    \noexpand\@setfontsize\noexpand\Huge
    {\the\@tempdima}{\the\@tempdimb}%
```

```
}%
```

We will now define some new font sizing commands. There are three variants for each standard command: \⟨*cmd*⟩r > \⟨*cmd*⟩rr > \⟨*cmd*⟩rrr. Every level is scaled proportionally in order to preserve the default value of the corresponding standard command.

```
\def\fontsize@pre{%
 \ctm@setlength{\abovedisplayskip}%
 {.8333\@tempdimb}{.1667\@tempdimb}{.5\@tempdimb}%
 \ctm@setlength{\abovedisplayshortskip}{\z@}{.25\@tempdimb}{\z@}%
 \ctm@setlength{\belowdisplayshortskip}%
 {.5\@tempdimb}{.25\@tempdimb}{.25\@tempdimb}%
 \setlength{\belowdisplayskip}{\abovedisplayskip}%
}
\def\fontsize@app{%
 \abovedisplayskip \the\abovedisplayskip%
 \abovedisplayshortskip \the\abovedisplayshortskip%
 \belowdisplayskip \the\belowdisplayskip%
 \belowdisplayshortskip \the\belowdisplayshortskip%
}
```

Definition of \smallr, \smallrr, \smallrrr.
Scale factor: 0,021875.

```
 \setlength{\@tempdima}{0.934375\@tempdimc}%
 \setlength{\@tempdimb}{\@tempb\@tempdima}%
 \fontsize@pre
 \edef\smallr{%
  \noexpand\@setfontsize\noexpand\smallr
  {\the\@tempdima}{\the\@tempdimb}%
  \fontsize@app
  \let\noexpand\@listi\noexpand\@listi@small
 }%
 \setlength{\@tempdima}{0.95625\@tempdimc}%
 \setlength{\@tempdimb}{\@tempb\@tempdima}%
 \fontsize@pre
 \edef\smallrr{%
  \noexpand\@setfontsize\noexpand\smallrr
  {\the\@tempdima}{\the\@tempdimb}%
  \fontsize@app
  \let\noexpand\@listi\noexpand\@listi@small
 }%
 \setlength{\@tempdima}{0.978125\@tempdimc}%
 \setlength{\@tempdimb}{\@tempb\@tempdima}%
 \fontsize@pre
 \edef\smallrrr{%
  \noexpand\@setfontsize\noexpand\smallrrr
  {\the\@tempdima}{\the\@tempdimb}%
  \fontsize@app
  \let\noexpand\@listi\noexpand\@listi@small
 }%
```

Definition of \footnotesizer, \footnotesizerr, \footnotesizerrr.
Scale factor: 0.01979.

```
\setlength{\@tempdima}{0.85313\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\fontsize@pre
\edef\footnotesizer{%
 \noexpand\@setfontsize\noexpand\footnotesizer
 {\the\@tempdima}{\the\@tempdimb}%
 \fontsize@app
 \let\noexpand\@listi\noexpand\@listi@footnotesize
}%
\setlength{\@tempdima}{0.87292\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\fontsize@pre
\edef\footnotesizerr{%
 \noexpand\@setfontsize\noexpand\footnotesizerr
 {\the\@tempdima}{\the\@tempdimb}%
 \fontsize@app
 \let\noexpand\@listi\noexpand\@listi@footnotesize
}%
\setlength{\@tempdima}{0.89271\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\fontsize@pre
\edef\footnotesizerrr{%
 \noexpand\@setfontsize\noexpand\footnotesizerrr
 {\the\@tempdima}{\the\@tempdimb}%
 \fontsize@app
 \let\noexpand\@listi\noexpand\@listi@footnotesize
}%
```

Definition of \scriptsizer, \scriptsizerr, \scriptsizerrr.
Scale factor: 0,0416675.

```
\setlength{\@tempdima}{0.7083375\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\scriptsizer{%
 \noexpand\@setfontsize\noexpand\scriptsizer
 {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{0.750005\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\scriptsizerr{%
 \noexpand\@setfontsize\noexpand\scriptsizerr
 {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{0.7916725\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\scriptsizerrr{%
 \noexpand\@setfontsize\noexpand\scriptsizerrr
 {\the\@tempdima}{\the\@tempdimb}%
}%
```

Definition of \tinyr, \tinyrr, \tinyrrr.
Scale factor: 0.0416675.

```
\setlength{\@tempdima}{0.5416675\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\tinyr{%
 \noexpand\@setfontsize\noexpand\tinyr
 {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{0.583335\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\tinyrr{%
 \noexpand\@setfontsize\noexpand\tinyrr
 {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{0.6250025\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\tinyrrr{%
 \noexpand\@setfontsize\noexpand\tinyrrr
 {\the\@tempdima}{\the\@tempdimb}%
}%
```

Definition of \normalsizer, \normalsizerr, \normalsizerrr.
Scale factor: 0.05.

```
\setlength{\@tempdima}{1.05\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
 \fontsize@pre
\edef\normalsizer{%
  \noexpand\@setfontsize\noexpand\normalsizer
  {\the\@tempdima}{\the\@tempdimb}%
 \fontsize@app
 \let\noexpand\@listi\noexpand\@listI
}%
\setlength{\@tempdima}{1.1\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
 \fontsize@pre
\edef\normalsizerr{%
  \noexpand\@setfontsize\noexpand\normalsizerr
  {\the\@tempdima}{\the\@tempdimb}%
 \fontsize@app
 \let\noexpand\@listi\noexpand\@listI
}%
\setlength{\@tempdima}{1.15\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
 \fontsize@pre
\edef\normalsizerrr{%
  \noexpand\@setfontsize\noexpand\normalsizerrr
  {\the\@tempdima}{\the\@tempdimb}%
 \fontsize@app
 \let\noexpand\@listi\noexpand\@listI
}%
```

Definition of \larger, \largerr, \largerrr.
Scale factor: 0,06.

```
\setlength{\@tempdima}{1.26\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\larger{%
  \noexpand\@setfontsize\noexpand\larger
  {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{1.32\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\largerr{%
  \noexpand\@setfontsize\noexpand\largerr
  {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{1.38\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\largerrr{%
  \noexpand\@setfontsize\noexpand\largerrr
  {\the\@tempdima}{\the\@tempdimb}%
}%
```

Definition of \Larger, \Largerr, \Largerrr.
Scale factor: 0,072.

```
\setlength{\@tempdima}{1.512\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\Larger{%
  \noexpand\@setfontsize\noexpand\Larger
  {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{1.584\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\Largerr{%
  \noexpand\@setfontsize\noexpand\Largerr
  {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{1.656\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\Largerrr{%
  \noexpand\@setfontsize\noexpand\Largerrr
  {\the\@tempdima}{\the\@tempdimb}%
}%
```

Definition of \LARGEr, \LARGErr, \LARGErrr.
Scale factor: 0,0865.

```
\setlength{\@tempdima}{1.8145\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\LARGEr{%
  \noexpand\@setfontsize\noexpand\LARGEr
  {\the\@tempdima}{\the\@tempdimb}%
```

```
}%
\setlength{\@tempdima}{1.901\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\LARGErr{%
  \noexpand\@setfontsize\noexpand\LARGErr
  {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{1.9875\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\LARGErrr{%
  \noexpand\@setfontsize\noexpand\LARGErrr
  {\the\@tempdima}{\the\@tempdimb}%
}%
```

Definition of \huger, \hugerr, \hugerrr.
Scale factor: 0,1035.

```
\setlength{\@tempdima}{2.1775\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\huger{%
  \noexpand\@setfontsize\noexpand\huger
  {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{2.281\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\hugerr{%
  \noexpand\@setfontsize\noexpand\hugerr
  {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{2.3845\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\hugerrr{%
  \noexpand\@setfontsize\noexpand\hugerrr
  {\the\@tempdima}{\the\@tempdimb}%
}%
```

Definition of \Huger, \Hugerr, \Hugerrr.
Scale factor: 0.123.

```
\setlength{\@tempdima}{2.611\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\Huger{%
  \noexpand\@setfontsize\noexpand\Huger
  {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{2.734\@tempdimc}%
\setlength{\@tempdimb}{\@tempb\@tempdima}%
\edef\Hugerr{%
  \noexpand\@setfontsize\noexpand\Hugerr
  {\the\@tempdima}{\the\@tempdimb}%
}%
\setlength{\@tempdima}{2.857\@tempdimc}%
```

```
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\Hugerrr{%
    \noexpand\@setfontsize\noexpand\Hugerrr
    {\the\@tempdima}{\the\@tempdimb}%
  }%
```

Definition of \HUGE, \HUGER, \HUGERR, \HUGERRR.
Scale factor: 0,149.

```
  \setlength{\@tempdima}{3.006\@tempdimc}%
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\HUGE{%
   \noexpand\@setfontsize\noexpand\HUGE
   {\the\@tempdima}{\the\@tempdimb}%
  }%
  \setlength{\@tempdima}{3.155\@tempdimc}%
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\HUGEr{%
   \noexpand\@setfontsize\noexpand\HUGEr
   {\the\@tempdima}{\the\@tempdimb}%
  }%
  \setlength{\@tempdima}{3.304\@tempdimc}%
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\HUGErr{%
   \noexpand\@setfontsize\noexpand\HUGErr
   {\the\@tempdima}{\the\@tempdimb}%
  }%
  \setlength{\@tempdima}{3.453\@tempdimc}%
  \setlength{\@tempdimb}{\@tempb\@tempdima}%
  \edef\HUGErrr{%
   \noexpand\@setfontsize\noexpand\HUGErrr
   {\the\@tempdima}{\the\@tempdimb}%
  }%

  \normalsize%
```

The KOMA files do not define \parindent and other lengths that are are strictly linked to the font rather than to the dimensions of the page and the text. So we first define \parindent which defaults to 1.5⟨*fontsize*⟩:

```
\setlength\parindent{\@fontsizeparindent}%
```

For the following lengths, all values are calculated proportionally to the main font. I took the bk11.clo file as a reference point. Note that the differences in the lengths affected by fontsize inside the bk*.clo files only concern \headheight, \headsep, \topskip and \footskip.

```
\setlength\smallskipamount{
  \fpeval{1.090909091\@tempdimb/4}pt \@plus 1\p@ \@minus 1\p@}%
\setlength\medskipamount{
  \fpeval{1.090909091\@tempdimb/2}pt \@plus 2\p@ \@minus 2\p@}%
\setlength\bigskipamount{
```

Table 2: The values of some specific lengths set by the package, in relation to some significant settings of the normal size. The values are expressed in pt and rounded to the second decimal place.

Normal size		10.00	10.10	10.20	10.30	10.40	10.50	10.60	10.70	10.80	10.90	11.00
\parindent		15.00	15.15	15.30	15.45	15.60	15.75	15.90	16.05	16.20	16.35	16.50
\smallskipamount	$\binom{plus}{minus}$ 1 pt)	2.73	2.75	2.78	2.81	2.84	2.86	2.89	2.92	2.95	2.97	3.00
\medskipamount	$\binom{plus}{minus}$ 2 pt)	5.45	5.51	5.56	5.62	5.67	5.73	5.78	5.84	5.89	5.95	6.00
\bigskipamount	$\binom{plus}{minus}$ 4 pt)	10.91	11.02	11.13	11.24	11.35	11.45	11.56	11.67	11.78	11.89	12.00

```
\fpeval{1.090909091\@tempdimb}pt \@plus 4\p@ \@minus 4\p@}%

\setlength{\@tempdimb}{\f@baselineskip}%
\ctm@setlength{\parsep}%
{.3333\@tempdimb}{.1667\@tempdimb}{.0833\@tempdimb}%
\ctm@setlength{\topsep}%
{.6667\@tempdimb}{.1667\@tempdimb}{.3333\@tempdimb}%
\@tempswafalse
\begingroup
\def\@list@extra{\aftergroup\@tempswatrue}%
\csname @listi\endcsname
\endgroup
\edef\@listi{\leftmargin\leftmargini
  \topsep \the\topsep
  \parsep \the\parsep
  \itemsep \parsep
  \if@tempswa\noexpand\@list@extra\fi
}%
\let\@listI\@listi
\ctm@setlength{\parsep}%
{.1667\@tempdimb}{.0833\@tempdimb}{.0833\@tempdimb}%
\ctm@setlength{\topsep}%
{.3333\@tempdimb}{.1667\@tempdimb}{.1667\@tempdimb}%
\@tempswafalse
\begingroup
\def\@list@extra{\aftergroup\@tempswatrue}%
\csname @listi\endcsname
\endgroup
\edef\@listi@small{\leftmargin\leftmargini
  \topsep \the\topsep
  \parsep \the\parsep
  \itemsep \parsep
  \if@tempswa\noexpand\@list@extra\fi
}%
\ctm@setlength{\parsep}%
{.125\@tempdimb}{.0625\@tempdimb}{.0625\@tempdimb}%
\ctm@setlength{\topsep}%
{.25\@tempdimb}{.125\@tempdimb}{.125\@tempdimb}%
\@tempswafalse
\begingroup
```

```
\def\@list@extra{\aftergroup\@tempswatrue}%
\csname @listi\endcsname
\endgroup
\edef\@listi@footnotesize{\leftmargin\leftmargini
  \topsep \the\topsep
  \parsep \the\parsep
  \itemsep \parsep
  \if@tempswa\noexpand\@list@extra\fi
}%
\ctm@setlength{\parsep}%
{.1667\@tempdimb}{.0833\@tempdimb}{.0833\@tempdimb}%
\ctm@setlength{\topsep}%
{.3333\@tempdimb}{.1667\@tempdimb}{.0833\@tempdimb}%
\@tempswafalse
\begingroup
\def\@list@extra{\aftergroup\@tempswatrue}%
\csname @listii\endcsname
\endgroup
\edef\@listii{\leftmargin\leftmarginii
  \labelwidth=\dimexpr \leftmargin-\labelsep
  \topsep \the\topsep
  \parsep \the\parsep
  \itemsep \parsep
  \if@tempswa\noexpand\@list@extra\fi
}%
\ctm@setlength{\topsep}%
{.1667\@tempdimb}{.0833\@tempdimb}{.0833\@tempdimb}%
\setlength{\partopsep}{\z@ \@plus\z@ \@minus\p@}%
\@tempswafalse
\begingroup
\def\@list@extra{\aftergroup\@tempswatrue}%
\csname @listiii\endcsname
\endgroup
\edef\@listiii{\leftmargin\leftmarginiii
  \labelwidth=\dimexpr \leftmargin-\labelsep
  \topsep \the\topsep
  \parsep \z@
  \partopsep \the\partopsep
  \itemsep \topsep
  \if@tempswa\noexpand\@list@extra\fi
}%
\@tempswafalse
\begingroup
\def\@list@extra{\aftergroup\@tempswatrue}%
\csname @listiv\endcsname
\endgroup
\edef\@listiv{\leftmargin\leftmarginiv
  \labelwidth=\dimexpr \leftmargin-\labelsep\relax
  \if@tempswa\noexpand\@list@extra\fi
}%
\@tempswafalse
```

```
  \begingroup
  \def\@list@extra{\aftergroup\@tempswatrue}%
  \csname @listv\endcsname
  \endgroup
  \edef\@listv{\leftmargin\leftmarginv
    \labelwidth=\dimexpr \leftmargin-\labelsep\relax
    \if@tempswa\noexpand\@list@extra\fi
  }%
  \@tempswafalse
  \begingroup
  \def\@list@extra{\aftergroup\@tempswatrue}%
  \csname @listvi\endcsname
  \endgroup
  \edef\@listvi{\leftmargin\leftmarginvi
    \labelwidth=\dimexpr \leftmargin-\labelsep\relax
    \if@tempswa\noexpand\@list@extra\fi
  }%
  \@listi
  \setlength{\@tempdimb}{\f@baselineskip}%
  \setlength{\footnotesep}{.555\@tempdimb}%
  \ctm@setlength{\skip\footins}%
  {.75\@tempdimb}{.3333\@tempdimb}{.1667\@tempdimb}%
  \ctm@setlength{\floatsep}%
  {\@tempdimb}{.1667\@tempdimb}{.1667\@tempdimb}%
  \ctm@setlength{\textfloatsep}%
  {1.6667\@tempdimb}{.1667\@tempdimb}{.3333\@tempdimb}%
  \setlength{\intextsep}{\floatsep}%
  \setlength{\dblfloatsep}{\floatsep}%
  \setlength{\dbltextfloatsep}{\textfloatsep}%
  \setlength{\@fptop}{\z@ \@plus 1fil}%
  \setlength{\@fpsep}{.6667\@tempdimb \@plus 2fil}%
  \setlength{\@fpbot}{\@fptop}%
  \setlength{\@dblfptop}{\@fptop}%
  \setlength{\@dblfpsep}{\@fpsep}%
  \setlength{\@dblfpbot}{\@fptop}%
  \ctm@setlength{\partopsep}{.2\@tempdimb}{.1\@tempdimb}{.1\@tempdimb}%
}
```

Excerpt from `scrfontsizes.sty` Here is a portion of the `scrfontsizes` package. It simply provides the `\generateclofile` command, without the KOMA related definitions. There are no reserved names for the `.clo` files. The name of the command is a new name for the KOMA `\generatefontfile`. In fact what the command generates is a `.clo` file.

```
\newcommand*{\generateclofile}[1]{%
 \kernel@ifnextchar [{\@generatefontfile{#1}}{\@@generatefontfile{#1}}%
}
\newcommand*{\@generatefontfile}{}
\def\@generatefontfile#1[#2]#3{%
 \begingroup
```

At least starting from version 3.31 of `scrfontsizes`, the code of the `\@generatefontfile`

and \@@generatefontfile commands sets the \parskip value to 0pt. We also use [#2] in place of [{#2}] for the optional argument of \changefontsize used immediately below.

```
 \setlength{\parskip}{0pt}%
 \changefontsize[#2]{#3}%
 \generate@fontfile{#1}{#3}%
 \endgroup
}
\newcommand\@@generatefontfile[2]{%
 \begingroup
 \setlength{\parskip}{0pt}%
 \changefontsize{#2}%
 \generate@fontfile{#1}{#2}%
 \endgroup
}
\newcommand*{\@@@generatefontfile}[2][]{}
\newcommand*{\@fontfile}{}
\let\@fontfile\@mainaux
\newcommand*{\generate@fontfile}[2]{%
 \def\@ind{\space\space}%
 \immediate\openout\@fontfile #1#2.clo
 \immediate\write\@fontfile{\@percentchar^^J%
  \@percentchar\space This is file `#1#2.clo', generated with^^J%
```

In the next three lines we only replace 'scrfontsizes' with 'fontsize' and 'Markus Kohm' with 'Ivan Valbusa', in order to get the correct information in the .log file.

```
  \@percentchar\space fontsize
  \csname ver@fontsize.\@pkgextension\endcsname,^^J%
  \@percentchar\space Copyright (c) Ivan Valbusa.^^J%
  \@percentchar^^J%
  \string\ProvidesFile{#1#2.clo}[%
  \the\year/\ifnum\month<10 0\fi\the\month/\ifnum\day<10 0\fi\the\day%
  \space generated font size class option (#2)]}%
 \immediate\write\@fontfile{%
  \string\def\string\normalsize{\@percentchar^^J%
   \@ind\string\@setfontsize\string\normalsize{\f@size}{\f@baselineskip}%
   \@percentchar^^J%
   \@ind\string\abovedisplayskip\space \the\abovedisplayskip^^J%
   \@ind\string\abovedisplayshortskip\space \the\abovedisplayshortskip^^J%
   \@ind\string\belowdisplayskip\space \the\belowdisplayskip^^J%
   \@ind\string\belowdisplayshortskip\space \the\belowdisplayshortskip^^J%
   \@ind\string\let\string\@listi\string\@listI^^J%
  }\@percentchar%
 }%
 \begingroup\small\@listi
 \immediate\write\@fontfile{%
  \string\def\string\small{\@percentchar^^J%
   \@ind\string\@setfontsize\string\small{\f@size}{\f@baselineskip}%
   \@percentchar^^J%
   \@ind\string\abovedisplayskip\space \the\abovedisplayskip^^J%
   \@ind\string\abovedisplayshortskip\space \the\abovedisplayshortskip^^J%
```

```
  \@ind\string\belowdisplayskip\space \the\belowdisplayskip^^J%
  \@ind\string\belowdisplayshortskip\space \the\belowdisplayshortskip^^J%
  \@ind\string\def\string\@listi{\@percentchar^^J%
   \@ind\@ind\string\leftmargin\string\leftmargini^^J%
   \@ind\@ind\string\topsep \the\topsep^^J%
   \@ind\@ind\string\parsep \the\parsep^^J%
   \@ind\@ind\string\itemsep \string\parsep^^J%
   \@ind}\@percentchar^^J%
 }\@percentchar%
}%
\endgroup
\begingroup\footnotesize\@listi
\immediate\write\@fontfile{%
 \string\def\string\footnotesize{\@percentchar^^J%
  \@ind\string\@setfontsize\string\footnotesize{\f@size}{\f@baselineskip}%
  \@percentchar^^J%
  \@ind\string\abovedisplayskip\space \the\abovedisplayskip^^J%
  \@ind\string\abovedisplayshortskip\space \the\abovedisplayshortskip^^J%
  \@ind\string\belowdisplayskip\space \the\belowdisplayskip^^J%
  \@ind\string\belowdisplayshortskip\space \the\belowdisplayshortskip^^J%
  \@ind\string\def\string\@listi{\@percentchar^^J%
   \@ind\@ind\string\leftmargin\string\leftmargini^^J%
   \@ind\@ind\string\topsep \the\topsep^^J%
   \@ind\@ind\string\parsep \the\parsep^^J%
   \@ind\@ind\string\itemsep \string\parsep^^J%
   \@ind}\@percentchar^^J%
 }\@percentchar%
}%
\endgroup
\begingroup\scriptsize
\immediate\write\@fontfile{%
 \string\def\string\scriptsize{%
  \string\@setfontsize\string\scriptsize{\f@size}{\f@baselineskip}%
 }\@percentchar%
}%
\endgroup
\begingroup\tiny
\immediate\write\@fontfile{%
 \string\def\string\tiny{%
  \string\@setfontsize\string\tiny{\f@size}{\f@baselineskip}%
 }\@percentchar%
}%
\endgroup
\begingroup\large
\immediate\write\@fontfile{%
 \string\def\string\large{%
  \string\@setfontsize\string\large{\f@size}{\f@baselineskip}%
 }\@percentchar%
}%
\endgroup
\begingroup\Large
```

```
\immediate\write\@fontfile{%
 \string\def\string\Large{%
  \string\@setfontsize\string\Large{\f@size}{\f@baselineskip}%
 }\@percentchar%
}%
\endgroup
\begingroup\LARGE
\immediate\write\@fontfile{%
 \string\def\string\LARGE{%
  \string\@setfontsize\string\LARGE{\f@size}{\f@baselineskip}%
 }\@percentchar%
}%
\endgroup
\begingroup\huge
\immediate\write\@fontfile{%
 \string\def\string\huge{%
  \string\@setfontsize\string\huge{\f@size}{\f@baselineskip}%
 }\@percentchar%
}%
\endgroup
\begingroup\Huge
\immediate\write\@fontfile{%
 \string\def\string\Huge{%
  \string\@setfontsize\string\Huge{\f@size}{\f@baselineskip}%
 }\@percentchar%
}%
\endgroup
```

The new sizing commands previously defined are recalled here and written in the .clo file if the `\generateclofile` command is used.

```
 \begingroup\tinyr
\immediate\write\@fontfile{%
 \string\def\string\tinyr{%
  \string\@setfontsize\string\tinyr{\f@size}{\f@baselineskip}%
 }\@percentchar%
}%
\endgroup
\begingroup\tinyrr
\immediate\write\@fontfile{%
 \string\def\string\tinyrr{%
  \string\@setfontsize\string\tinyrr{\f@size}{\f@baselineskip}%
 }\@percentchar%
}%
\endgroup
\begingroup\tinyrrr
\immediate\write\@fontfile{%
 \string\def\string\tinyrrr{%
  \string\@setfontsize\string\tinyrrr{\f@size}{\f@baselineskip}%
 }\@percentchar%
}%
 \endgroup
```

```
 \begingroup\scriptsizer
 \immediate\write\@fontfile{%
  \string\def\string\scriptsizer{%
   \string\@setfontsize\string\scriptsizer{\f@size}{\f@baselineskip}%
  }\@percentchar%
 }%
 \endgroup
 \begingroup\scriptsizerr
 \immediate\write\@fontfile{%
  \string\def\string\scriptsizerr{%
   \string\@setfontsize\string\scriptsizerr{\f@size}{\f@baselineskip}%
  }\@percentchar%
 }%
\endgroup
 \begingroup\scriptsizerrr
 \immediate\write\@fontfile{%
  \string\def\string\scriptsizerrr{%
   \string\@setfontsize\string\scriptsizerrr{\f@size}{\f@baselineskip}%
  }\@percentchar%
 }%
 \endgroup
\def\fs@appnormal{
 \@percentchar^^J%
 \@ind\string\abovedisplayskip\space \the\abovedisplayskip^^J%
 \@ind\string\abovedisplayshortskip\space \the\abovedisplayshortskip^^J%
 \@ind\string\belowdisplayskip\space \the\belowdisplayskip^^J%
 \@ind\string\belowdisplayshortskip\space \the\belowdisplayshortskip^^J%
 \@ind\string\let\string\@listi\string\@listI^^J%
}
\def\fs@appsmall{%
 \@percentchar^^J%
 \@ind\string\abovedisplayskip\space \the\abovedisplayskip^^J%
 \@ind\string\abovedisplayshortskip\space \the\abovedisplayshortskip^^J%
 \@ind\string\belowdisplayskip\space \the\belowdisplayskip^^J%
 \@ind\string\belowdisplayshortskip\space \the\belowdisplayshortskip^^J%
 \@ind\string\def\string\@listi{\@percentchar^^J%
  \@ind\@ind\string\leftmargin\string\leftmargini^^J%
  \@ind\@ind\string\topsep \the\topsep^^J%
  \@ind\@ind\string\parsep \the\parsep^^J%
  \@ind\@ind\string\itemsep \string\parsep^^J%
  \@ind}\@percentchar^^J%
}
 \begingroup\smallr\@listi
 \immediate\write\@fontfile{%
  \string\def\string\smallr{\@percentchar^^J%
   \@ind\string\@setfontsize\string\smallr{\f@size}{\f@baselineskip}%
   \fs@appsmall%
  }\@percentchar%
 }%
 \endgroup
 \begingroup\smallrr\@listi
```

```
\immediate\write\@fontfile{%
 \string\def\string\smallrr{\@percentchar^^J%
  \@ind\string\@setfontsize\string\smallrr{\f@size}{\f@baselineskip}%
  \fs@appsmall%
 }\@percentchar%
}%
\endgroup
\begingroup\smallrrr\@listi
\immediate\write\@fontfile{%
 \string\def\string\smallrrr{\@percentchar^^J%
  \@ind\string\@setfontsize\string\smallrrr{\f@size}{\f@baselineskip}%
  \fs@appsmall%
 }\@percentchar%
}%
\endgroup
\begingroup\footnotesizer\@listi
\immediate\write\@fontfile{%
 \string\def\string\footnotesizer{\@percentchar^^J%
  \@ind\string\@setfontsize\string\footnotesizer{\f@size}{\f@baselineskip}%
  \fs@appsmall%
 }\@percentchar%
}%
\endgroup
\begingroup\footnotesizerr\@listi
\immediate\write\@fontfile{%
 \string\def\string\footnotesizerr{\@percentchar^^J%
 \@ind\string\@setfontsize\string\footnotesizerr{\f@size}{\f@baselineskip}%
  \fs@appsmall%
 }\@percentchar%
}%
\endgroup
\begingroup\footnotesizerrr\@listi
\immediate\write\@fontfile{%
 \string\def\string\footnotesizerrr{\@percentchar^^J%
 \@ind\string\@setfontsize\string\footnotesizerrr{\f@size}{\f@baselineskip}%
  \fs@appsmall%
 }\@percentchar%
}%
\endgroup
\begingroup\normalsizer
\immediate\write\@fontfile{%
\string\def\string\normalsizer{\@percentchar^^J%
  \@ind\string\@setfontsize\string\normalsizer{\f@size}{\f@baselineskip}%
 \fs@appnormal%
  }\@percentchar%
}%
\endgroup
\begingroup\normalsizerr
\immediate\write\@fontfile{%
 \string\def\string\normalsizerr{\@percentchar^^J%
  \@ind\string\@setfontsize\string\normalsizerr{\f@size}{\f@baselineskip}%
```

```
  \fs@appnormal%
 }\@percentchar%
}%
\endgroup
\begingroup\normalsizerrr
\immediate\write\@fontfile{%
\string\def\string\normalsizerrr{\@percentchar^^J%
  \@ind\string\@setfontsize\string\normalsizerrr{\f@size}{\f@baselineskip}%
  \fs@appnormal%
 }\@percentchar%
}%
\endgroup
\begingroup\larger
\immediate\write\@fontfile{%
  \string\def\string\larger{%
    \string\@setfontsize\string\larger{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
\endgroup
\begingroup\largerr
\immediate\write\@fontfile{%
  \string\def\string\largerr{%
    \string\@setfontsize\string\largerr{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
\endgroup
\begingroup\largerrr
\immediate\write\@fontfile{%
  \string\def\string\largerrr{%
    \string\@setfontsize\string\largerrr{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
\endgroup
\begingroup\Large
\immediate\write\@fontfile{%
  \string\def\string\Large{%
    \string\@setfontsize\string\Large{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
\endgroup
\begingroup\Larger
\immediate\write\@fontfile{%
  \string\def\string\Larger{%
    \string\@setfontsize\string\Larger{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
\endgroup
\begingroup\Largerr
\immediate\write\@fontfile{%
  \string\def\string\Largerr{%
    \string\@setfontsize\string\Largerr{\f@size}{\f@baselineskip}%
```

```
  }\@percentchar%
}%
\endgroup
\begingroup\Largerrr
\immediate\write\@fontfile{%
  \string\def\string\Largerrr{%
    \string\@setfontsize\string\Largerrr{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
\endgroup
\begingroup\LARGEr
\immediate\write\@fontfile{%
  \string\def\string\LARGEr{%
    \string\@setfontsize\string\LARGEr{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
\endgroup
\begingroup\LARGErr
\immediate\write\@fontfile{%
  \string\def\string\LARGErr{%
    \string\@setfontsize\string\LARGErr{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
\endgroup
\begingroup\LARGErrr
\immediate\write\@fontfile{%
  \string\def\string\LARGErrr{%
    \string\@setfontsize\string\LARGErrr{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
\endgroup
\begingroup\huger
\immediate\write\@fontfile{%
  \string\def\string\huger{%
    \string\@setfontsize\string\huger{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
\endgroup
\begingroup\hugerr
\immediate\write\@fontfile{%
  \string\def\string\hugerr{%
    \string\@setfontsize\string\hugerr{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
\endgroup
\begingroup\hugerrr
\immediate\write\@fontfile{%
  \string\def\string\hugerrr{%
    \string\@setfontsize\string\hugerrr{\f@size}{\f@baselineskip}%
  }\@percentchar%
}%
```

```
\endgroup
 \begingroup\Huger
 \immediate\write\@fontfile{%
   \string\def\string\Huger{%
     \string\@setfontsize\string\Huger{\f@size}{\f@baselineskip}%
   }\@percentchar%
 }%
 \endgroup
 \begingroup\Hugerr
 \immediate\write\@fontfile{%
   \string\def\string\Hugerr{%
     \string\@setfontsize\string\Hugerr{\f@size}{\f@baselineskip}%
   }\@percentchar%
 }%
 \endgroup
 \begingroup\Hugerrr
 \immediate\write\@fontfile{%
   \string\def\string\Hugerrr{%
     \string\@setfontsize\string\Hugerrr{\f@size}{\f@baselineskip}%
   }\@percentchar%
 }%
\endgroup
  \begingroup\HUGE
 \immediate\write\@fontfile{%
  \string\def\string\HUGE{%
   \string\@setfontsize\string\HUGE{\f@size}{\f@baselineskip}%
  }\@percentchar%
 }%
 \endgroup
  \begingroup\HUGEr
 \immediate\write\@fontfile{%
  \string\def\string\HUGEr{%
   \string\@setfontsize\string\HUGEr{\f@size}{\f@baselineskip}%
  }\@percentchar%
 }%
 \endgroup
 \begingroup\HUGErr
 \immediate\write\@fontfile{%
  \string\def\string\HUGErr{%
   \string\@setfontsize\string\HUGErr{\f@size}{\f@baselineskip}%
  }\@percentchar%
 }%
 \endgroup
 \begingroup\HUGErrr
 \immediate\write\@fontfile{%
  \string\def\string\HUGErrr{%
   \string\@setfontsize\string\HUGErrr{\f@size}{\f@baselineskip}%
  }\@percentchar%
 }%
\endgroup
```

```
\immediate\write\@fontfile{\string\normalsize}%
```

The new lengths previously defined are recalled here and written in the .clo file if the `\generateclofile` command is used.

```
\begingroup
\immediate\write\@fontfile{%
  \string\setlength{\string\parindent}{\the\parindent}^^J%
  \string\setlength{\string\headheight}{\the\headheight}^^J%
  \string\setlength{\string\smallskipamount}{\the\smallskipamount}^^J%
  \string\setlength{\string\medskipamount}{\the\medskipamount}^^J%
  \string\setlength{\string\bigskipamount}{\the\bigskipamount}^^J%
}%
\endgroup

 \immediate\write\@fontfile{%
  \string\setlength{\string\footnotesep}{\the\footnotesep}\@percentchar
 }%
 \immediate\write\@fontfile{%
 \string\setlength{\string\skip\string\footins}{\the\skip\footins}\@percentchar
 }%
 \immediate\write\@fontfile{%
  \string\setlength{\string\floatsep}{\the\floatsep}\@percentchar
 }%
 \immediate\write\@fontfile{%
  \string\setlength{\string\textfloatsep}{\the\textfloatsep}\@percentchar
 }%
 \immediate\write\@fontfile{%
  \string\setlength{\string\intextsep}{\the\intextsep}\@percentchar
 }%
 \immediate\write\@fontfile{%
  \string\setlength{\string\dblfloatsep}{\the\dblfloatsep}\@percentchar
 }%
 \immediate\write\@fontfile{%
 \string\setlength{\string\dbltextfloatsep}{\the\dbltextfloatsep}\@percentchar
 }%
 \immediate\write\@fontfile{%
  \string\setlength{\string\@fptop}{\the\@fptop}\@percentchar
 }%
 \immediate\write\@fontfile{%
  \string\setlength{\string\@fpsep}{\the\@fpsep}\@percentchar
 }%
 \immediate\write\@fontfile{%
  \string\setlength{\string\@fpbot}{\the\@fpbot}\@percentchar
 }%
 \immediate\write\@fontfile{%
  \string\setlength{\string\@dblfptop}{\the\@dblfptop}\@percentchar
 }%
 \immediate\write\@fontfile{%
  \string\setlength{\string\@dblfpsep}{\the\@dblfpsep}\@percentchar
 }%
 \immediate\write\@fontfile{%
```

```
 \string\setlength{\string\@dblfpbot}{\the\@dblfpbot}\@percentchar
}%
\immediate\write\@fontfile{%
 \string\setlength{\string\partopsep}{\the\partopsep}\@percentchar
}%
\begingroup
\@listi
\immediate\write\@fontfile{%
 \string\def\string\@listi{\@percentchar^^J%
  \@ind\string\leftmargin\string\leftmargini^^J%
  \@ind\string\topsep \the\topsep^^J%
  \@ind\string\parsep \the\parsep^^J%
  \@ind\string\itemsep \string\parsep^^J%
 }\@percentchar
}%
\endgroup
\immediate\write\@fontfile{%
 \string\let\string\@listI\string\@listi
}%
\begingroup
\@listii
\immediate\write\@fontfile{%
 \string\def\string\@listii{\@percentchar^^J%
  \@ind\string\leftmargin\string\leftmarginii^^J%
  \@ind\string\labelwidth\string\leftmarginii^^J%
  \@ind\string\advance\string\labelwidth-\string\labelsep^^J%
  \@ind\string\topsep \the\topsep^^J%
  \@ind\string\parsep \the\parsep^^J%
  \@ind\string\itemsep \string\parsep^^J%
 }\@percentchar
}%
\endgroup
\begingroup
\@listiii
\immediate\write\@fontfile{%
 \string\def\string\@listiii{\@percentchar^^J%
  \@ind\string\leftmargin\string\leftmarginiii^^J%
  \@ind\string\labelwidth\string\leftmarginiii^^J%
  \@ind\string\advance\string\labelwidth-\string\labelsep^^J%
  \@ind\string\topsep \the\topsep^^J%
  \@ind\string\parsep \the\parsep^^J%
  \@ind\string\partopsep \the\partopsep^^J%
  \@ind\string\itemsep \string\topsep^^J%
 }\@percentchar
}%
\endgroup
\begingroup
\@listiv
\immediate\write\@fontfile{%
 \string\def\string\@listiv{\@percentchar^^J%
  \@ind\string\leftmargin\string\leftmarginiv^^J%
```

```
   \@ind\string\labelwidth\string\leftmarginiv^^J%
   \@ind\string\advance\string\labelwidth-\string\labelsep^^J%
  }\@percentchar
 }%
 \endgroup
 \begingroup
 \@listv
 \immediate\write\@fontfile{%
  \string\def\string\@listv{\@percentchar^^J%
   \@ind\string\leftmargin\string\leftmarginv^^J%
   \@ind\string\labelwidth\string\leftmarginv^^J%
   \@ind\string\advance\string\labelwidth-\string\labelsep^^J%
  }\@percentchar
 }%
 \endgroup
 \begingroup
 \@listvi
 \immediate\write\@fontfile{%
  \string\def\string\@listvi{\@percentchar^^J%
   \@ind\string\leftmargin\string\leftmarginvi^^J%
   \@ind\string\labelwidth\string\leftmarginvi^^J%
   \@ind\string\advance\string\labelwidth-\string\labelsep^^J%
  }\@percentchar
 }%
 \endgroup
 \immediate\write\@fontfile{%
 \string\@listi^^J%
 \string\endinput^^J%
 \@percentchar^^J%
 \@percentchar\space End of file `#1#2.clo'.
}
\immediate\closeout\@fontfile
}
\@onlypreamble\generateclofile
```

We will now define some internal commands to test the package functions. Since version 0.6 I have been using the LaTeX3 solution provided by Enrico Gregorio in place of the old macros based on `etoolbox` (Lehman and Wright 2019).

```
\newcommand{\sampletext}[1]{\def\fs@sampletext{#1}}
\def\fs@sampletext{%
  As any dedicated reader can clearly see, the Ideal of practical reason
  is a representation of, as far as I know, the things in themselves; as
  I have shown elsewhere, the phenomena should only be used as a canon for
  our understanding. The paralogisms of practical reason are what first give
  rise to the architectonic of practical reason. As will easily be shown in
  the next section, reason would thereby be made to contradict, in view of
  these considerations, the Ideal of practical reason, yet the manifold
  depends on the phenomena.}

\ExplSyntaxOn

```

```
\NewDocumentCommand{\printsamples}
{
 m
 m
 O{tiny,tinyr,tinyrr,tinyrrr,
   scriptsize,scriptsizer,scriptsizerr,scriptsizerrr,
   footnotesize,footnotesizer,footnotesizerr,footnotesizerrr,
   small,smallr,smallrr,smallrrr,
   normalsize,normalsizer,normalsizerr,normalsizerrr,
   large,larger,largerr,largerrr,
   Large,Larger,Largerr,Largerrr,
   LARGE,LARGEr,LARGErr,LARGErrr,
   huge,huger,hugerr,hugerrr,
   Huge,Huger,Hugerr,Hugerrr,
   HUGE,HUGEr,HUGErr,HUGErrr,
  }
}
{
 \fontsize_printsamples:nnn { #1 } { #2 } { #3 }
}

% helper function to print the rounded sizes
\cs_new:Nn \__fontsize_round_f:n
{
 \fp_eval:n { round(\use:c{f@#1},1) } pt
}

% the main function
\cs_new_protected:Nn \fontsize_printsamples:nnn
{
 % group to bound the font changes
 \group_begin:
 % set the size
 \changefontsize[#1]{#2}\selectfont
 % header
 \subsubsection*{Text~samples~for~\symbol{92}normalsize~#2~on~#1}
 % map through the desired sizes
 \clist_map_inline:nn { #3 }
 {
  \use:c{##1}                              % set the font size
  \noindent\texttt{\symbol{92}##1}:~       % print the font size command
  \__fontsize_round_f:n {size}~on~         % print the size in points
  \__fontsize_round_f:n {baselineskip}.~   % print the baseline skip in points
  \fs@sampletext                           % sample text
  \par\addvspace{\medskipamount}           % finish up
 }
 % close the group
 \group_end:
}
\ExplSyntaxOff
```

References

Adriaens, Hendri (2014), *The* xkeyval *package. Extension of the keyval package*, version 2.7a, http://www.ctan.org/pkg/xkeyval (visited on 07/05/2020).

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Change History

v0.1
General: First public release 1
v0.2
General: Updated documentation and code comments. 1
v0.3
General: Added definitions of some lengths omitted from KOMA-script files: \smallskiamount, \medskiamount, \bigskiamount. Redefined \parindent and \headheight. 1
v0.4
General: Maintenance release. Updated documentation. 1
v0.5
General: New font sizing commands from \normalsize up. New command \printsamples. Updated documentation. 1
v0.6
General: New font sizing commands from \small down. Updated documentation. 1
v0.7
General: New font sizing commands \HUGE, \HUGEr, \HUGErr, \HUGErrr. Changed default name for the .clo file to fontsize. Updated documentation. 1
v0.8
General: Corrected a bug occuring when the line spacing is grater than 163.84 pt. 1
v0.8.1
General: New \sampletext command. Some lengths are now defined via xfp package. Removed \headheight definition. Updated documentation. 1
v0.8.2
General: New \parindent option. Updated documentation. 1
v0.8.3
General: Corrected a bug in the \parindent option. Updated documentation. 1
v0.8.4
General: Maintenance release. 1
v0.8.5
General: Updated documentation. . . . 1

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